QUAKELOGIC

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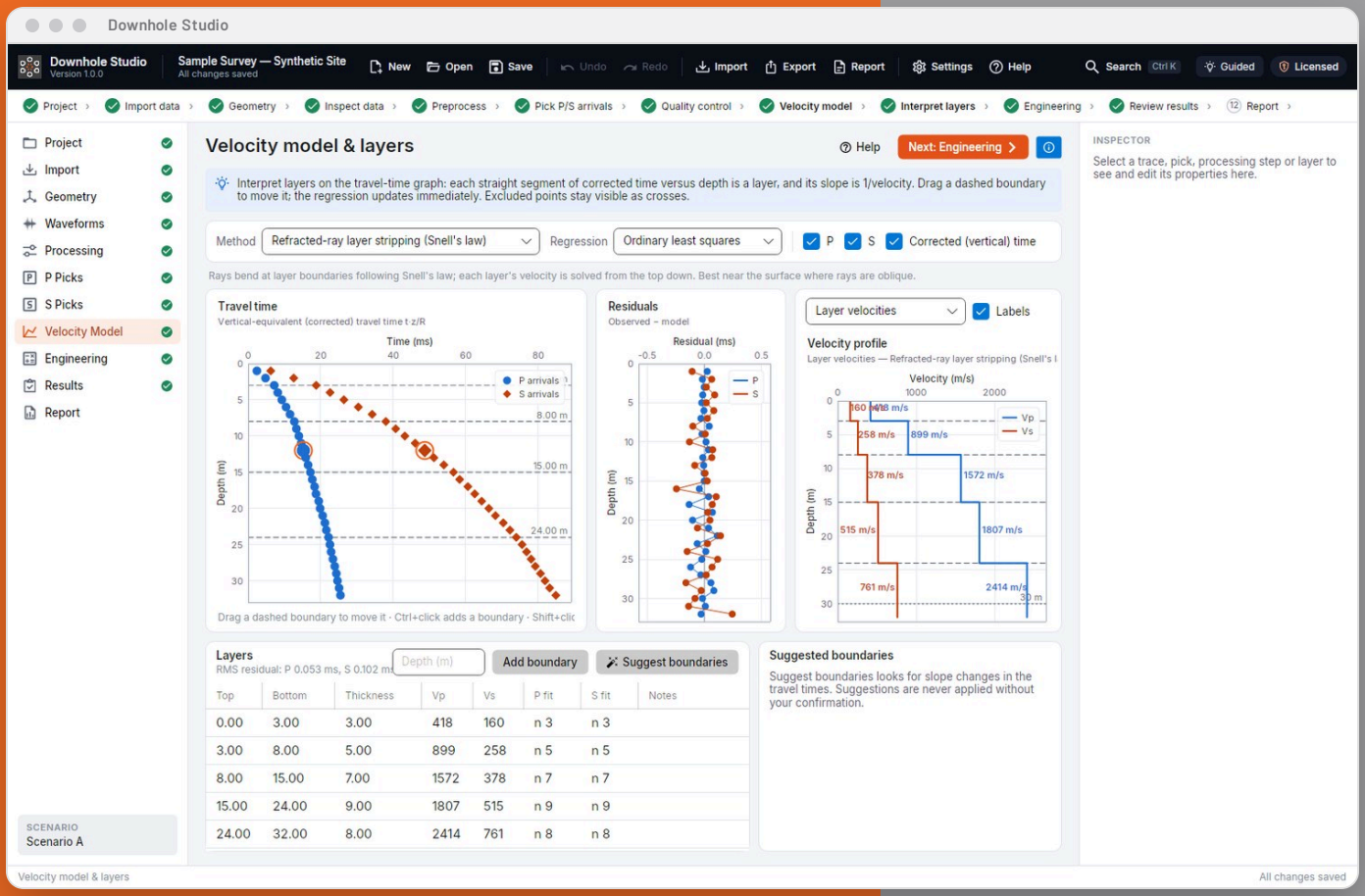
METHODS & VALIDATION

DOWNHOLE STUDIO

EQUATIONS · ASSUMPTIONS · INDEPENDENT CHECKS

Every calculation with its assumptions and limitations, and the validation report for this version.

DOWNHOLE STUDIO 1.0.0 · WINDOWS 10 / 11



TECHNICAL REFERENCE

Methods and scientific validation

DOWNHOLE STUDIO 1.0.0 · FOR REVIEWING ENGINEERS

How Downhole Studio calculates every result, the assumptions behind each method and the independent checks that verify them. Downhole Studio supports the established downhole workflow described in ASTM D7400/D7400M but does not certify compliance with any standard; results require professional review.

1 Scientific validation

2 Signal processing

3 First-arrival picking

4 Reference geophone and trigger correction

5 Travel-time geometry and velocity methods

6 Layer interpretation

7 Vs30, engineering parameters and density

8 Quality control

9 Downhole Studio — scientific validation report

PART 1 · SCIENTIFIC VALIDATION

Scientific correctness takes precedence over producing an answer. When a value cannot be computed reliably, Downhole Studio states why instead of producing a number.

Method

Every release is checked by an automated validation run, reproduced in full in Part 3 of this document. Every row lists the input, the expected value, the value Downhole Studio calculated, the difference, the tolerance and the result. Expected values come from three independent sources:

1. **Hand calculation** for each formula (straight-ray distance, vertical time, interval velocity, Vs30, Poisson's ratio, G, E, K, density estimate, unit conversions, trigger correction).
2. **An independent implementation** for ray tracing: the engine solves Snell's law by bisection on the ray parameter; the synthetic generator minimises travel time over interface crossing points (Fermat's principle) with Newton iterations. They agree to 10^{-9} s.
3. **Known synthetic earth models**: five SEG-2 datasets (clean, noisy, missing traces, trigger errors, reversed horizontal polarity) are processed exactly as a user would (import, geometry, reference picks, trigger correction, automatic picks, layers, refracted-ray velocities) and the pick times, layer Vp and Vs and Vs30 are compared with the model.

Current result

87 of 87 checks pass (version 1.0.0). Highlights: S-pick RMS error 0.09–0.21 ms on clean data and 0.94 ms on noisy data; layer Vs within 4 % (8 % noisy) of the model; Vs30 within 0.2–1.2 % of the true 367.6 m/s.

Known limitations

- Straight-ray methods (pseudo- and true interval, vertical-time correction) are biased where rays are oblique below slower layers; the true-interval method is only validated where depth/offset ≥ 4. Use the refracted-ray method near the surface.
- The refracted-ray model ignores head waves and dipping layers.
- Automatic picks on emergent onsets tend to be slightly late (0.1–0.3 ms in the clean tests); zero-phase filtering before picking can make them early. Review picks on lightly processed data.
- Suggested layer boundaries can be misplaced on noisy data; the analyst decides.
- Empirical density relations were calibrated on rocks.
- Validation uses synthetic data. Before production use, QuakeLogic should add comparisons with field surveys interpreted independently (e.g. with established software or published case histories) under the same reporting format.

Relation to standards

The workflow follows established downhole practice (ASTM D7400/D7400M: source offset correction, vertical travel-time correction, interval and layer velocities, P and S identification with opposite shots). The software does not reproduce the standard and does not certify compliance; the user remains responsible for professional judgment and applicable requirements.

PART 2 · METHODS

Units: all calculations use SI units (m, s, m/s, kg/m^3, Pa) in double precision. Display units are applied only in the user interface and in exports; converted values are never fed back into calculations.

Signal processing

All operations return new arrays; recorded samples are never modified.

Order of operations

channel-map polarity → enabled pipeline steps in order → per-trace polarity → per-trace static shift. Display gain is applied only when drawing.

Mean removal and linear detrend

Mean: $x'_i = x_i - \bar{x}$ (Kahan-compensated sum). Detrend: least-squares line $a + b \cdot i$ removed, with the index centred so the normal equations are diagonal.

Taper

Hann (raised-cosine) ramp $w = \tfrac{1}{2}(1 - \cos(\pi i / m))$ over $m = \lfloor f \cdot N \rfloor$ samples at each end ($f \le 0.5$). Limitation: arrivals inside the tapered portion are attenuated; the UI warns about this.

Butterworth filters

Analogue prototype poles $p_k = \exp(i\pi(2k + N + 1)/(2N))$, $k = 0 \ldots N-1$. Low-pass $s = \omega_c \cdot p$, high-pass $s = \omega_c / p$, band-pass from the roots of $s^2 - p \cdot B \cdot s + \omega_0^2 = 0$ ($B = \omega_2 - \omega_1$, $\omega_0^2 = \omega_1 \omega_2$). Corners are pre-warped, $\omega = 2fs \cdot \tan(\pi f / fs)$, and mapped with the bilinear transform $z = (2fs + s)/(2fs - s)$. Zeros at $s = \infty$ map to $z = -1$, zeros at $s = 0$ to $z = +1$. Conjugate poles are paired into second-order sections and the gain is normalised at DC (low-pass), Nyquist (high-pass) or the pre-warped geometric centre (band-pass). The corner frequency is the −3 dB point.

Causal filtering uses direct form II transposed. Zero-phase filtering runs the cascade forward and backward over an odd reflection of length $3(2 \cdot \text{sections} + 1)$ with steady-state initial conditions (for a biquad: $z_2 = b_2 - a_2 G$, $z_1 = b_1 - a_1 G + z_2$ for unit input, G the DC gain), giving $|H|^2$ and zero phase. Results match SciPy 1.18 `butter(..., output='sos')`, `sosfilt` and `sosfiltfilt` to 1e−8 (tests: `FilterTests`).

Limitation: zero-phase filters are acausal — energy can appear slightly before a sharp onset; causal filters delay the waveform. Pick first breaks on raw or lightly processed data.

Notch

Bilinear notch with exact −3 dB bandwidth f_0/Q: $\beta = \tan(\omega_0/(2Q))$, $g = 1/(1 + \beta)$, $b = g \cdot [1, -2\cos\omega_0, 1]$, $a = [1, -2g\cos\omega_0, 2g - 1]$ (equivalent to SciPy `iirnotch`). Optional integer harmonics below Nyquist.

Normalisation, balance and AGC

Peak: divide by $\max|x|$. RMS: divide by $\sqrt{(\Sigma x^2 / N)}$. Balance: RMS in a user window = 1. AGC: each sample divided by the RMS in a centred window (prefix sums; samples whose RMS is below 10^{-6} of the global RMS are set to zero). AGC alters relative amplitudes and is a display aid.

Time shift

Whole-sample shifts are exact. Fractional shifts multiply the spectrum of a zero-padded copy (length ≥ $2N + |\text{shift}|$) by $\exp(-2\pi i \cdot f \cdot \tau)$ with a real Nyquist bin, so the waveform does not wrap.

Resampling

Windowed-sinc interpolation (Blackman window, ±16 zero crossings) normalised by the kernel sum. When downsampling, the kernel cut-off is 0.9 × the new Nyquist frequency (anti-aliasing).

Spectrum

Single-sided amplitude $|X(f)|\cdot 2/N$ of a Hann-windowed trace with window-gain compensation. Dominant frequency = largest amplitude above the minimum frequency; bandwidth = contiguous band around the peak above a fraction of the peak (−3 dB: $1/\sqrt{2}$; −6 dB: 1/2). FFT: radix-2 for powers of two, Bluestein's chirp-z for other lengths (exact for any N).

Envelope

$|x + iH\{x\}|$ from the FFT analytic signal.

Cross-correlation

$r(\tau) = \Sigma\, a(t)\cdot b(t+\tau) / \sqrt{(\Sigma a^2 \cdot \Sigma b^2)}$ via FFT, peak refined by parabolic interpolation. The reported coefficient for windowed analyses is the Pearson correlation of the window of *a* with *b* shifted by the best lag. Positive lag ⇒ *b* is later than *a*. Polarity-insensitive mode maximises $|r|$.

Horizontal rotation

$R = X\cos\theta + Y\sin\theta$, $T = -X\sin\theta + Y\cos\theta$ (θ from X towards Y). Applied only when the user chooses an angle; the angle is stored per depth and reported.

First-arrival picking

All picks are stored in raw record time (seconds after the recorded trigger). Corrections are applied in the analysis, never written into the pick.

AIC picker (default)

Maeda (1985) direct AIC on the search window $x[0\ldots N-1]$:

$$AIC(k) = k\cdot \ln \mathrm{var}(x[0\ldots k]) + (N-k-1)\cdot \ln \mathrm{var}(x[k+1\ldots N-1])$$

The onset is the global minimum. Because a window that also contains the decay of the arrival has a second variance change (signal → quiet), the window is ended shortly after the largest amplitude (peak + max(8, ¼ of the distance from the window start)). Sharpness = rise of AIC 5 % of the window either side of the minimum relative to 15 % of the AIC range (0–1).

STA/LTA

Allen (1978): ratio of the mean energy in a trailing short window ending at the sample to that in a long window immediately before it. The onset is the start of the rise (last local minimum of the ratio) before the first crossing of the trigger level.

Modified energy ratio

Han, Wong & Liu (2009): $ER(i) = \Sigma_{i\ldots i+L}\, x^2 \,/\, \Sigma_{i-L\ldots i}\, x^2$, $MER(i) = (ER(i)\cdot|x(i)|)^3$; onset at the maximum of MER.

Search window

From the record start (including pre-trigger samples) or a user time, to a user time or the record end. With neighbour prediction, the search is restricted to ±2× the prediction half-width around the time predicted from a straight line of time versus slant distance through the nearest picked depths (processed top to bottom).

Signal-to-noise ratio

RMS amplitude in [t, t + signal window) divided by RMS in [t − noise window, t).

Confidence

Weighted mean of the available factors (weights renormalised when a factor is missing):

FACTOR	WEIGHT	SCORE
SNR	0.35	$(\log_{10}\mathrm{SNR} - \log_{10} 2)/(\log_{10} 12 - \log_{10} 2)$, clamped to 0–1
Onset sharpness	0.20	picker sharpness
Algorithm agreement	0.20	1 if the closest other algorithm is within max(3 samples, 0.3 ms P / 1 ms S), decreasing linearly to 0 at 5× that
Neighbour consistency	0.15	$\exp(-(\Delta/(2\cdot\mathrm{limit}))^2)$, Δ = difference from the neighbour prediction, limit = QC residual limit
Opposite-polarity agreement (S)	0.10	−r between S^+ and the aligned S^-

High ≥ 0.75, Medium ≥ 0.45, Low otherwise (thresholds configurable). Each factor carries a sentence explaining its score; the inspector shows them.

S-wave traces

S^+ and S^- shots of the chosen horizontal component (X, Y or rotated by the per-depth angle) are stacked per polarity when repeated, the S^- shot is shifted into the S^+ trigger time base (difference of trigger corrections) and the half-difference $(S^+ - S^-)/2$ is formed. Energy that reverses with the source direction (SH waves) is kept; energy common to both shots (P waves, noise) cancels.

Manual picks

Manual picks are snapped to the nearest sample of the trace (picks are not finer than the data). Re-running the automatic picker never replaces a manual pick unless requested; the new suggestion is stored next to it and can be accepted or restored.

Reference geophone and trigger correction

A reference (spy) geophone at the source records the moment of impact in every record. Its onset t_ref is picked automatically (AIC over the first 50 ms, including pre-trigger samples) or by hand.

MODE	CORRECTION ADDED TO RAW TIMES	USE
None	0	trigger is reliable
Subtract reference onset	−t_ref + t_expected	times measured from the impact; t_expected = travel time from source to reference geophone (normally ≈ 0)
Align to median	−(t_ref − median(t_ref))	remove shot-to-shot jitter but keep the common trigger offset

Raw time, reference onset, correction and corrected time are kept and reported for every depth. A record without a reference onset cannot be corrected; its arrivals are excluded from the velocity analysis with the reason shown (never silently used uncorrected). When repeated shots are stacked, or S^- is combined with S^+, the traces are first aligned by the difference of their trigger corrections.

QC flags reference onsets that differ from the median by more than the limit (default 0.5 ms): as a likely problem when no correction is applied, as a note when the correction removes it, and for review when the deviation exceeds four times the limit (possible reference mispick).

Travel-time geometry and velocity methods

Geometry

z = receiver depth below ground (measured depth minus collar height when depths are measured from the collar), z_s = source depth below ground, x = horizontal source offset (separate for P and S).

- Slant distance: $R = \sqrt{((z - z_s)^2 + x^2)}$
- Corrected (vertical-equivalent) time: $t_v = t \cdot (z - z_s)/R$
- Average velocity to a receiver: $V = R/t$

Assumption for t_v: a straight ray in a medium whose velocity does not vary along the ray. Near the surface below a slow layer, rays bend; use the refracted-ray method there.

Layer regression (default)

In each interpreted layer, fit $t_v = a + z/V$ to the receivers inside the layer (receivers exactly on a boundary are used by both layers): $V = 1/\text{slope}$. Ordinary least squares, or Huber M-estimation by IRLS with the scale re-estimated each iteration as $1.4826 \cdot \text{MAD}$ ($k = 1.345$ by default). Reported: V, $\pm 1\sigma$ from the slope standard error ($\sigma_V = \sigma_\text{slope}/\text{slope}^2$), R^2, n, RMS residual.

Pseudo-interval

$V = (z_2 - z_1)/(t_v{,}2 - t_v{,}1)$ between adjacent usable receivers. Sensitive to small pick errors.

True interval (straight rays)

$V = (R_2 - R_1)/(t_2 - t_1)$ with trigger-corrected times. Assumes straight rays through a uniform medium between source and receiver; biased below slower layers where rays are oblique (validated only where $z/x \geq 4$).

For the interval methods the layer velocity spans the first and last usable receivers inside the layer (the same formula over the layer).

Refracted-ray layer stripping

Rays obey Snell's law through horizontal layers (constant ray parameter $p = \sin\theta_j / v_j$). For a receiver at depth z with offset x the ray parameter solves $X(p) = \Sigma\, h_j \cdot p \cdot v_j / \sqrt{(1 - p^2 v_j^2)} = x$ (monotonic in p; bisection), and the travel time is $t = \Sigma\, h_j / (v_j \sqrt{(1 - p^2 v_j^2)})$. Layers are solved from the top down: each layer's velocity minimises the squared misfit between observed (trigger-corrected) times and ray times for the receivers it contains, holding the layers above fixed (log-spaced grid 10–10 000 m/s refined by golden-section search). Head waves are not modelled; a layer without receivers blocks the layers below. The ray-tracer agrees with an independent Fermat-principle minimiser to 10^{-9} s (validation report §2).

Excluded observations

Picks can be excluded (or a depth or record excluded). Excluded observations stay visible (crosses on the plots), are listed with the reason in the report and audit record, and are not used in any fit.

Layer interpretation

The layer model is a list of layer tops (the first at 0 m); each layer's bottom is the next top or the model bottom (deepest usable receiver unless set). Editing operations — add, move, delete, merge, split boundaries, set densities — are pure functions recorded as undoable edits.

Suggested boundaries

Optional and never applied without confirmation:

1. Optimal piecewise-linear segmentation of corrected time versus depth for $k = 1 \ldots 8$ segments by dynamic programming (each segment ≥ 3 receivers, cost = Σ squared residuals of a line fit).
2. k is chosen by the Bayesian information criterion $n \cdot \ln(\text{SSE}/n) + (3k - 1) \cdot \ln n$.
3. The largest segmentation not above the BIC choice in which adjacent segments differ in velocity by at least 15 % is kept (smaller changes are not resolvable layers).

4. Each boundary is placed at the intersection of the two adjacent fitted lines when it falls between the bracketing receivers, otherwise mid-way between them.

Each suggestion states the evidence (velocity above and below). With noisy picks, suggestions can be misplaced; the analyst decides, ideally guided by borehole logs.

Vs30, engineering parameters and density

Time-averaged velocity and Vs30

$V(z_t, z_b) = (z_b - z_t) / \Sigma\, h_i/V_i$ over the layer portions inside $[z_t, z_b]$; $Vs30 = V(0, 30\ m)$. If the deepest usable S measurement is shallower than z_b, the value is **not calculated** and the reason is shown ("Insufficient measured depth for direct Vs30 calculation"). Only if the user explicitly allows extrapolation is the deepest layer's Vs extended to z_b; the value is then labelled EXTRAPOLATED in the application, tables, plots and the report. The first layer's velocity is assumed to apply from the ground surface; a note is raised when the shallowest measurement is deeper than 2 m.

Elastic properties (isotropic, small strain)

- Poisson's ratio $\nu = (Vp^2 - 2Vs^2)/(2(Vp^2 - Vs^2))$
- Shear modulus $G = \rho Vs^2$
- Young's modulus $E = 2G(1 + \nu)$
- Bulk modulus $K = \rho(Vp^2 - 4Vs^2/3)$
- Constrained modulus $M = \rho Vp^2$

The equivalent forms $E = \rho Vs^2(3Vp^2 - 4Vs^2)/(Vp^2 - Vs^2)$, $K = E/(3(1 - 2\nu))$, $G = E/(2(1 + \nu))$ and $K + 4G/3 = M$ are checked numerically in the tests.

Safety rules (values withheld and flagged, never corrected): $Vs \geq Vp$; $Vp \approx Vs$ (ν undefined); $K \leq 0$ ($Vp/Vs \leq 2/\sqrt{3}$); $E \leq 0$; missing density (moduli not calculated). $\nu < 0$ is reported as unusual. When $Vp > 1400$ m/s and $Vp/Vs > 3$, a note explains that below the water table Vp reflects the pore fluid.

Density

Never assumed. Sources:

- **Measured** — per layer, entered by the user.
- **Constant** — one value entered by the user.
- **Estimated** — only when explicitly selected; labelled ESTIMATED:
  - Gardner, Gardner & Gregory (1974): $\rho = 1.74 \cdot Vp^{0.25}$ (g/cm³, Vp in km/s); calibrated on brine-saturated sedimentary rocks, 1.5–6.1 km/s.
  - Brocher (2005) Nafe–Drake fit: $\rho = 1.6612Vp - 0.4721Vp^2 + 0.0671Vp^3 - 0.0043Vp^4 + 0.000106Vp^5$; 1.5–8.5 km/s. Outside the calibration range a warning is attached. Both relations were developed for rocks and are approximate for soils.

Units

US customary display uses exact factors: 1 ft = 0.3048 m, 1 lbm = 0.45359237 kg (1 pcf = 16.0184634 kg/m³), standard gravity 9.80665 m/s² (1 psi = 6894.757 Pa, 1 ksf = 47 880.26 Pa).

Quality control

Checks flag; they never change data.

CHECK	SEVERITY	RULE (DEFAULTS, CONFIGURABLE)
Low SNR	red < 3, yellow < 6	pick SNR
Weak onset	yellow	sharpness < 0.35
Low confidence	yellow	automatic pick scored Low (with the reasons)
Unreviewed automatic pick	note	automatic pick not accepted
Large residual	yellow > limit, red > 2×limit	P 0.5 ms, S 1.5 ms, once layers are interpreted

CHECK	SEVERITY	RULE (DEFAULTS, CONFIGURABLE)		
Arrival earlier at a deeper receiver	red	corrected time decreases with slant distance		
Unusual velocity jump	yellow	adjacent pseudo-interval velocities differ by > 2.5×		
Invalid interval	yellow	$\Delta t \le 0$ or $\Delta z \le 0$		
No opposite-polarity confirmation	yellow	S^+ or S^- missing		
Same polarity in S^+ and S^-	red	correlation > 0 around the S pick		
Poor opposite-shot correlation	yellow		r	< 0.6
Possible incorrect component	yellow	horizontal energy > 3× vertical around a P pick, or the reverse for S		
Trigger timing	red/yellow/note	reference onset deviates from the median by > 0.5 ms (see Reference geophone and trigger correction)		
Missing reference onset	red	correction enabled but no onset		
Poor regression	yellow	$R^2 < 0.95$ with ≥ 3 points		
Layer without velocity	yellow	reason given		
Elastic consistency	red/yellow	see Vs30, engineering parameters and density		
Vs30 unavailable / extrapolated	yellow			

Verdict. *Incomplete* when no data, no depths, fewer than two usable P or S arrivals, fewer than half the depths with S, or no layer velocities. *Review recommended* when any red flag, more than 20 % of picks with yellow pick flags, Vs30 unavailable or extrapolated, or unreviewed automatic picks. Otherwise *Good*. The verdict is advisory and does not certify compliance with any standard.

PART 3 · VALIDATION REPORT

Generated 2026-10-10 23:42 UTC by Downhole Studio 1.0.0. 87 of 87 checks passed.

Expected values come from hand calculation, from an independent implementation (Fermat-principle ray timing in the synthetic generator) or from the known synthetic earth model. Tolerances are absolute unless marked %.

1. Formulas (hand calculation)

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
Slant distance R = $\sqrt{((z-zs)^2+x^2)}$	z=4, zs=0, x=3 m	5.0000 m	5.0000 m	0.000000	0.000000 m	PASS	3-4-5 triangle
Vertical time t·z/R	t=10 ms, z=4, x=3	0.008000 s	0.008000 s	0.000000	0.000000 s	PASS	0.010×4/5
Interval velocity $(R_2-R_1)/(t_2-t_1)$	R 5→10 m, t 10→20 ms	500.0000 m/s	500.0000 m/s	0.000000	0.000000 m/s	PASS	5/0.010
Vs30 = 30/Σ(h/V)	5 m@200, 10 m@300, 15 m@500	339.6226 m/s	339.6226 m/s	0.000000	0.000001 m/s	PASS	30/0.0883333
Poisson $\nu=(Vp^2-2Vs^2)/(2(Vp^2-Vs^2))$	Vp 1500, Vs 300	0.479167	0.479167	-0.000000	0.000000	PASS	2.07e6/4.32e6
Shear modulus $G=\rho Vs^2$	ρ 1900, Vs 300	171000000.0 Pa	171000000.0 Pa	0.000000	0.001000 Pa	PASS	1900×300^2
Young's modulus $E=2G(1+\nu)$	as above	505875000.0 Pa	505875000.0 Pa	0.000000	1.0000 Pa	PASS	$\rho Vs^2(3Vp^2-4Vs^2)/(Vp^2-Vs^2)$
Bulk modulus $K=\rho(Vp^2-4Vs^2/3)$	as above	4047000000.0 Pa	4047000000.0 Pa	0.000000	1.0000 Pa	PASS	1900×(2.25e6–1.2e5)
Gardner density	Vp 2000 m/s	2069.2 kg/m³	2069.2 kg/m³	0.020380	0.100000 kg/m³	PASS	1.74×2^0.25 g/cm³
ft → m	1 ft	0.304800 m	0.304800 m	0.000000	0.000000 m	PASS	exact definition
pcf → kg/m³	1 lbm/ft³	16.0185 kg/m³	16.0185 kg/m³	0.000000	0.000000 kg/m³	PASS	0.45359237/0.3048³
ksf → Pa	1 kip/ft²	47880.3 Pa	47880.3 Pa	0.000000	0.000001 Pa	PASS	4448.2216/0.09290304
psi → Pa	1 lbf/in²	6894.8 Pa	6894.8 Pa	0.000000	0.000001 Pa	PASS	$4.4482216/0.0254^2$
Trigger correction t − t_ref	t=15.0 ms, t_ref=1.2 ms	0.013800 s	0.013800 s	0.000000	0.000000 s	PASS	15.0 − 1.2 ms

2. Ray tracing (independent implementations)

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
S travel time, Snell (engine) vs Fermat (generator)	z=1 m, x=2 m	0.013975 s	0.013975 s	0.000000	0.000000 s	PASS	Fermat minimisation
S travel time, Snell (engine) vs Fermat (generator)	z=2.5 m, x=2 m	0.020010 s	0.020010 s	0.000000	0.000000 s	PASS	Fermat minimisation
S travel time, Snell (engine) vs Fermat (generator)	z=6 m, x=3 m	0.033627 s	0.033627 s	0.000000	0.000000 s	PASS	Fermat minimisation
S travel time, Snell (engine) vs Fermat (generator)	z=12 m, x=2 m	0.049108 s	0.049108 s	0.000000	0.000000 s	PASS	Fermat minimisation
S travel time, Snell (engine) vs Fermat (generator)	z=20 m, x=4 m	0.067138 s	0.067138 s	0.000000	0.000000 s	PASS	Fermat minimisation

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
S travel time, Snell (engine) vs Fermat (generator)	z=31 m, x=2 m	0.083058 s	0.083058 s	0.000000	0.000000 s	PASS	Fermat minimisation
S travel time, Snell (engine) vs Fermat (generator)	z=31 m, x=8 m	0.085081 s	0.085081 s	0.000000	0.000000 s	PASS	Fermat minimisation

3. Synthetic dataset — Clean

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
P pick RMS error vs true onset	32 picks	0.000000 s	0.000065 s	0.000065	0.000250 s	PASS	generator onset times
S pick RMS error vs true onset	32 picks	0.000000 s	0.000192 s	0.000192	0.000500 s	PASS	generator onset times
Vs 0–3 m (refracted ray)	true layer model	160.0000 m/s	159.2894 m/s	-0.710563	4 %	PASS	synthetic model
Vp 0–3 m (refracted ray)	true layer model	420.0000 m/s	418.8895 m/s	-1.1105	6 %	PASS	synthetic model
Vs 3–8 m (refracted ray)	true layer model	260.0000 m/s	259.3576 m/s	-0.642424	4 %	PASS	synthetic model
Vp 3–8 m (refracted ray)	true layer model	900.0000 m/s	899.3360 m/s	-0.663983	6 %	PASS	synthetic model
Vs 8–15 m (refracted ray)	true layer model	380.0000 m/s	379.2540 m/s	-0.745987	4 %	PASS	synthetic model
Vp 8–15 m (refracted ray)	true layer model	1600.0 m/s	1588.8 m/s	-11.2003	6 %	PASS	synthetic model
Vs 15–24 m (refracted ray)	true layer model	520.0000 m/s	516.7169 m/s	-3.2831	4 %	PASS	synthetic model
Vp 15–24 m (refracted ray)	true layer model	1850.0 m/s	1841.8 m/s	-8.2353	6 %	PASS	synthetic model
Vs 24–32 m (refracted ray)	true layer model	760.0000 m/s	765.1997 m/s	5.1997	4 %	PASS	synthetic model
Vp 24–32 m (refracted ray)	true layer model	2400.0 m/s	2401.3 m/s	1.2645	6 %	PASS	synthetic model
Vs30	true layer model	367.6279 m/s	366.6223 m/s	-1.0056	2 %	PASS	30/Σ(h/Vs) of the model

3. Synthetic dataset — Noisy

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
P pick RMS error vs true onset	32 picks	0.000000 s	0.000627 s	0.000627	0.000750 s	PASS	generator onset times
S pick RMS error vs true onset	32 picks	0.000000 s	0.000943 s	0.000943	0.001500 s	PASS	generator onset times
Vs 0–3 m (refracted ray)	true layer model	160.0000 m/s	155.6718 m/s	-4.3282	8 %	PASS	synthetic model
Vp 0–3 m (refracted ray)	true layer model	420.0000 m/s	410.5518 m/s	-9.4482	12 %	PASS	synthetic model
Vs 3–8 m (refracted ray)	true layer model	260.0000 m/s	258.8661 m/s	-1.1339	8 %	PASS	synthetic model
Vp 3–8 m (refracted ray)	true layer model	900.0000 m/s	871.1064 m/s	-28.8936	12 %	PASS	synthetic model
Vs 8–15 m (refracted ray)	true layer model	380.0000 m/s	373.5752 m/s	-6.4248	8 %	PASS	synthetic model

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
Vp 8–15 m (refracted ray)	true layer model	1600.0 m/s	1595.7 m/s	-4.3345	12 %	PASS	synthetic model
Vs 15–24 m (refracted ray)	true layer model	520.0000 m/s	516.1414 m/s	-3.8586	8 %	PASS	synthetic model
Vp 15–24 m (refracted ray)	true layer model	1850.0 m/s	1783.1 m/s	-66.8894	12 %	PASS	synthetic model
Vs 24–32 m (refracted ray)	true layer model	760.0000 m/s	764.6861 m/s	4.6861	8 %	PASS	synthetic model
Vp 24–32 m (refracted ray)	true layer model	2400.0 m/s	2112.9 m/s	-287.1074	12 %	PASS	synthetic model
Vs30	true layer model	367.6279 m/s	363.1626 m/s	-4.4653	5 %	PASS	30/Σ(h/Vs) of the model

3. Synthetic dataset — MissingTraces

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
P pick RMS error vs true onset	31 picks	0.000000 s	0.000084 s	0.000084	0.000250 s	PASS	generator onset times
S pick RMS error vs true onset	32 picks	0.000000 s	0.000198 s	0.000198	0.000500 s	PASS	generator onset times
Vs 0–3 m (refracted ray)	true layer model	160.0000 m/s	159.2894 m/s	-0.710563	4 %	PASS	synthetic model
Vp 0–3 m (refracted ray)	true layer model	420.0000 m/s	418.8895 m/s	-1.1105	6 %	PASS	synthetic model
Vs 3–8 m (refracted ray)	true layer model	260.0000 m/s	259.4429 m/s	-0.557104	4 %	PASS	synthetic model
Vp 3–8 m (refracted ray)	true layer model	900.0000 m/s	892.8007 m/s	-7.1993	6 %	PASS	synthetic model
Vs 8–15 m (refracted ray)	true layer model	380.0000 m/s	377.9514 m/s	-2.0486	4 %	PASS	synthetic model
Vp 8–15 m (refracted ray)	true layer model	1600.0 m/s	1604.9 m/s	4.8840	6 %	PASS	synthetic model
Vs 15–24 m (refracted ray)	true layer model	520.0000 m/s	520.8402 m/s	0.840190	4 %	PASS	synthetic model
Vp 15–24 m (refracted ray)	true layer model	1850.0 m/s	1830.3 m/s	-19.7116	6 %	PASS	synthetic model
Vs 24–32 m (refracted ray)	true layer model	760.0000 m/s	754.7071 m/s	-5.2929	4 %	PASS	synthetic model
Vp 24–32 m (refracted ray)	true layer model	2400.0 m/s	2379.8 m/s	-20.2067	6 %	PASS	synthetic model
Vs30	true layer model	367.6279 m/s	366.4951 m/s	-1.1328	2 %	PASS	30/Σ(h/Vs) of the model
Missing P record reported, not invented	record 6:P omitted	1.0000 depths	1.0000 depths	0.000000	0.000000 depths	PASS	dataset definition

3. Synthetic dataset — TriggerErrors

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
P pick RMS error vs true onset	32 picks	0.000000 s	0.000057 s	0.000057	0.000250 s	PASS	generator onset times
S pick RMS error vs true onset	32 picks	0.000000 s	0.000087 s	0.000087	0.000500 s	PASS	generator onset times
Vs 0–3 m (refracted ray)	true layer model	160.0000 m/s	160.7160 m/s	0.716024	4 %	PASS	synthetic model

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
Vp 0–3 m (refracted ray)	true layer model	420.0000 m/s	421.6210 m/s	1.6210	6 %	PASS	synthetic model
Vs 3–8 m (refracted ray)	true layer model	260.0000 m/s	258.4251 m/s	-1.5749	4 %	PASS	synthetic model
Vp 3–8 m (refracted ray)	true layer model	900.0000 m/s	900.6537 m/s	0.653698	6 %	PASS	synthetic model
Vs 8–15 m (refracted ray)	true layer model	380.0000 m/s	380.9839 m/s	0.983860	4 %	PASS	synthetic model
Vp 8–15 m (refracted ray)	true layer model	1600.0 m/s	1578.3 m/s	-21.7148	6 %	PASS	synthetic model
Vs 15–24 m (refracted ray)	true layer model	520.0000 m/s	515.2687 m/s	-4.7313	4 %	PASS	synthetic model
Vp 15–24 m (refracted ray)	true layer model	1850.0 m/s	1838.8 m/s	-11.1929	6 %	PASS	synthetic model
Vs 24–32 m (refracted ray)	true layer model	760.0000 m/s	762.8393 m/s	2.8393	4 %	PASS	synthetic model
Vp 24–32 m (refracted ray)	true layer model	2400.0 m/s	2368.5 m/s	-31.5470	6 %	PASS	synthetic model
Vs30	true layer model	367.6279 m/s	367.1078 m/s	-0.520155	2 %	PASS	30/Σ(h/Vs) of the model

3. Synthetic dataset — ReversedPolarityS

CHECK	INPUT	EXPECTED	ACTUAL	DIFFERENCE	TOLERANCE	RESULT	EXPECTED FROM
P pick RMS error vs true onset	32 picks	0.000000 s	0.000071 s	0.000071	0.000250 s	PASS	generator onset times
S pick RMS error vs true onset	32 picks	0.000000 s	0.000208 s	0.000208	0.000500 s	PASS	generator onset times
Vs 0–3 m (refracted ray)	true layer model	160.0000 m/s	159.0698 m/s	-0.930174	4 %	PASS	synthetic model
Vp 0–3 m (refracted ray)	true layer model	420.0000 m/s	418.8895 m/s	-1.1105	6 %	PASS	synthetic model
Vs 3–8 m (refracted ray)	true layer model	260.0000 m/s	259.9139 m/s	-0.086134	4 %	PASS	synthetic model
Vp 3–8 m (refracted ray)	true layer model	900.0000 m/s	899.3360 m/s	-0.663983	6 %	PASS	synthetic model
Vs 8–15 m (refracted ray)	true layer model	380.0000 m/s	378.6849 m/s	-1.3151	4 %	PASS	synthetic model
Vp 8–15 m (refracted ray)	true layer model	1600.0 m/s	1590.0 m/s	-9.9691	6 %	PASS	synthetic model
Vs 15–24 m (refracted ray)	true layer model	520.0000 m/s	517.7024 m/s	-2.2976	4 %	PASS	synthetic model
Vp 15–24 m (refracted ray)	true layer model	1850.0 m/s	1826.0 m/s	-23.9547	6 %	PASS	synthetic model
Vs 24–32 m (refracted ray)	true layer model	760.0000 m/s	758.8371 m/s	-1.1629	4 %	PASS	synthetic model
Vp 24–32 m (refracted ray)	true layer model	2400.0 m/s	2421.4 m/s	21.3970	6 %	PASS	synthetic model
Vs30	true layer model	367.6279 m/s	366.4205 m/s	-1.2074	2 %	PASS	30/Σ(h/Vs) of the model

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